

Mayank Manohar — Manma AI Knowledge Base

A structured personal and professional knowledge document for retrieval-augmented generation (RAG). This document is intended to help Manma answer visitor questions about Mayank accurately and consistently.

1. Quick Profile

Name	Mayank Manohar
Role	Full Stack Developer
Education	B.Tech, Dr. B. C. Roy Engineering College
Expected graduation	2027
Location	India
Primary stack	MERN, Docker, AWS, GSAP
AI / GenAI	LLMs, RAG, embeddings, LangChain, Google Gemini, Qdrant
Portfolio	MAYANK.OS

2. About Mayank

Mayank is an engineering student and Full Stack Developer focused on building complete, usable web applications. His main development background is the MERN ecosystem, with additional experience in Docker, AWS, GSAP, REST APIs, databases, and applied AI/GenAI. He is particularly interested in connecting frontend interfaces, backend systems, deployment, and AI capabilities into practical products.

His current portfolio is built around the MAYANK.OS concept. The portfolio presents his projects, technical skills, profile, contact information, and Manma, an AI portfolio assistant that uses retrieval-augmented generation.

3. Education

Dr. B. C. Roy Engineering College

B.Tech Engineering

Expected passing year: 2027

4. Technical Skills

Area	Technologies
Languages	Java, JavaScript, SQL
Frontend	React.js, HTML5, CSS3, Tailwind CSS, GSAP, Responsive Design
Backend	Node.js, Express.js, REST APIs, JWT Authentication
Databases	MongoDB, MySQL / SQL, Qdrant, vector database concepts
AI / GenAI	Large Language Models (LLMs), RAG, LangChain, embeddings, Google Gemini
Machine Learning	Python, NLP, NumPy, Pandas, scikit-learn, TensorFlow, Streamlit
DevOps / Cloud	Docker, AWS (EC2, ECR), Vercel, Render

Area	Technologies
Tools	Git, GitHub, VS Code, Postman

5. Featured Projects

Manma — AI Portfolio Assistant

Year: 2026 **Category:** AI Engineering / Full Stack

Description: An AI-powered portfolio assistant that helps visitors understand Mayank's skills, projects, experience, and technical background. It uses retrieval-augmented generation so answers can be grounded in Mayank's own knowledge base.

Technologies: React, Node.js, Express, LangChain, Google Gemini, RAG, embeddings, Qdrant

Key details:

- Personal knowledge system for portfolio information.
- Uses document chunking and embeddings.
- Uses Qdrant as the vector database for retrieval.
- Uses LangChain for the AI/RAG workflow.
- Uses Google Gemini for generating responses.
- Integrated into the MAYANK.OS portfolio through a custom chat interface.
- Backend exposes a POST /manma API endpoint.
- Backend is deployed on Render.

Live: <https://mayank-os-portfolio.vercel.app>

GitHub: <https://github.com/manohar-mayank/manma-backend>

Interview AI Strategy Builder

Year: 2026 **Category:** AI Engineering / Full Stack

Description: An AI-powered application designed to help users prepare for technical interviews through structured interview strategies and preparation guidance.

Technologies: React, Node.js, Express, MongoDB, REST APIs, LLM API

Key details:

- AI-powered interview preparation workflow.
- React frontend with Node.js/Express backend.
- REST API architecture.
- Authentication and application flow.
- Deployed full-stack application.

Live: <https://interview-ai-strategy-builder.vercel.app>

GitHub: <https://github.com/manohar-mayank/interview-ai-strategy-builder>

RasoiMitra — Smart Recipe Companion

Year: 2025 **Category:** Full Stack / Frontend

Description: A recipe discovery application that helps users find recipes based on ingredients and explore nutrition, instructions, and related food content.

Technologies: JavaScript, Node.js, Spoonacular API, REST API, HTML, CSS

Key details:

- Ingredient-based recipe discovery.
- External food API integration.
- Nutrition and recipe information.
- Responsive user interface.
- Animated landing experience.

Live: <https://rasoi-mitra-web-app.vercel.app>

GitHub: <https://github.com/manohar-mayank/RasoiMitra-Web-App>

Emotion Detection System

Year: 2026 **Category:** AI / Machine Learning

Description: A machine-learning application that analyzes text and predicts emotional categories through an interactive Streamlit interface.

Technologies: Python, NLP, Machine Learning, Pandas, NumPy, scikit-learn, Streamlit

Key details:

- Text-based emotion classification.
- NLP preprocessing workflow.
- Machine-learning model pipeline.
- Interactive Streamlit interface.
- Deployed ML application.

Live: <https://real-time-emotions-detection.streamlit.app>

GitHub: <https://github.com/manohar-mayank/real-time-emotion-detection>

SkyCast — Weather Application

Year: 2025 **Category:** Frontend

Description: A responsive weather application that provides real-time weather information using the OpenWeatherMap API.

Technologies: HTML, CSS, JavaScript, OpenWeatherMap API

Key details:

- Real-time weather data.
- OpenWeatherMap API integration.
- Dynamic weather information.
- Responsive interface.

Live: <https://sky-cast-eight-alpha.vercel.app>

GitHub: <https://github.com/manohar-mayank/SkyCast>

Employee Payroll Management System

Year: 2025 **Category:** Java / Backend

Description: A Java desktop application for managing employee payroll, salary history, deductions, and department-level reporting.

Technologies: Java, Java Swing, MySQL, JFreeChart

Key details:

- Employee CRUD operations.
- Payroll and salary management.
- MySQL database persistence.
- Department-wise reporting and charts.
- CSV and payslip functionality.
- Leave deduction and hike-related payroll functionality.

GitHub: <https://github.com/manohar-mayank/Employee-Payroll-System>

6. Manma — Technical Knowledge

Manma is the AI assistant integrated into Mayank's portfolio. Its purpose is to make it easier for visitors to ask questions about Mayank instead of reading every portfolio section manually.

Architecture:

- Frontend: React portfolio interface.
- API: Node.js / Express POST /manma endpoint.
- LLM: Google Gemini.
- Orchestration: LangChain.
- Knowledge retrieval: RAG.
- Representation: text chunks and embeddings.
- Vector database: Qdrant.
- Deployment: frontend on Vercel and backend on Render.

Manma Knowledge Updating

When Mayank changes important portfolio information, the knowledge-base source document should be updated and the changed content should be re-chunked, embedded, and inserted or updated in Qdrant. The goal is for retrieval to use the latest information rather than relying on stale chunks.

7. Certifications, Programs & Experience

- **Webskitters Technology Solutions Pvt. Ltd., Kolkata:** Mayank worked as a Full Stack Development (AI) Intern at Webskitters Technology Solutions Pvt. Ltd., Kolkata, during August 2026. The internship was a 60-hour Summer Internship and Industry Immersion program focused on Full Stack Development using AI. During the internship, he worked on a MERN Stack E-Commerce Management System using React.js, Node.js, Express.js, and MongoDB. The project included modules for product catalog management, shopping cart functionality, and order workflows.
- **Google Student Ambassador:** Mayank participated in the Google Student Ambassador program at Dr. B. C. Roy Engineering College during November–December 2025. He completed program tasks and campus activities and received a Google kit and associated recognition.
- **CTTC Training:** Completed a one-month training program covering NLP, Machine Learning, NumPy, Pandas, scikit-learn and related topics. Training project: Emotion Detection System.
- **Python:** Has a Basics of Python certificate.

8. Deployment & Engineering Experience

- Has deployed frontend applications using Vercel.
- Has deployed backend services using Render.
- Has worked with environment variables for frontend/backend communication.
- Has worked with CORS configuration for frontend-to-backend API requests.
- Has worked with Docker and AWS concepts including EC2 and ECR.
- Has experience debugging deployment issues such as incorrect ports, CORS origins, environment variables, and API connection problems.

9. Current Development Focus

Mayank is focused on strengthening his full-stack engineering profile while building practical AI-integrated applications. His current direction combines MERN development, backend/API engineering, deployment, and applied GenAI/RAG systems.

He is interested in building substantial full-stack and AI projects rather than only small tutorial applications. His portfolio is intended to demonstrate complete product development: interface, backend, database, AI integration, deployment, and documentation.

10. Career Profile

Mayank is preparing for software development opportunities, with a focus on full-stack development and roles where he can work with modern web technologies and AI-integrated systems. He is particularly interested in off-campus opportunities and wants his GitHub, LinkedIn, resume, and portfolio to demonstrate real project work.

11. Contact & Professional Links

Email	manoharmayank33@gmail.com
Phone	+91 9472582351
GitHub	https://github.com/manohar-mayank
LinkedIn	https://www.linkedin.com/in/manohar-mayank
Portfolio	https://mayank-os-portfolio.vercel.app

12. Accuracy Rules for Manma

- Do not claim that Mayank has used LangGraph; he has not used it.
- Do not claim RasoiMitra uses React; it does not.
- Do not invent project features, certifications, companies, internships, or technologies that are not documented here.
- When information is not available in the knowledge base, Manma should say that the information is not currently available.
- Prefer specific project evidence over generic claims about expertise.
- For questions about contact details, use the contact section in this document.

End of Mayank Manohar knowledge base. This document is structured for chunking and retrieval in a RAG pipeline.